Project-based Internship Agreement

Intern Host: Expando World Limited

Intern Students: 莫羡瑜 Mo Xianyu

both agree to engage in an internship for the research and development project under ITC STEM Internship Program. Both agree on the following items:

Start Date	2 Dec 2025	**Internship Topic**	LLM-based Proactive AR Agent
End Date	2 Feb 2026	**Working Days**	Full-time from Monday to Friday
Postion	LLM multi-modal AI developer	**Stipend**	$11490 each month provided by STEM Internship Program
Probation Period	28 days	**Clearance Time**	After internship ends and the program issues the stipend

1. Duties of Intern Host and Students

1.1 Both will agree to the roles and clear processes maintained by both parties. In the case of internship everybody's role needs to be clarified.

1.2 Both will respond timely to messages and discuss regularly to make progress towards the completion of the project research and development.

1.3 Both will keep one another informed of any planned vacations or absences as well as changes in his or her personal circumstances that might impact on the scheduled work progress. Unplanned absences or delays should be discussed as soon as possible and arrangements should be made, to catch up lost time.

2. Tasks and Duties of Intern Students

2.1 The intern students should make their research and development plan and follow the instructions of the intern host to make continuous progress towards the expected goals and milestones (see attachment). By the end of the internship, students should finish their research and development work and deliver the expected outcomes to meet their goals.

2.2 The intern students should timely record their working process and report their working progress every working day on their working daily provided by the intern host. They should also put all their work on the Github repository (for code) and online document (for any other else) provided by the intern host. Their work should be reducible and reproducible including detailed instructions and process step by step. They should deploy their work on the online server or cloud platform provided by the intern host and make them runnable and workable as expected.

2.3 The intern students also attend and present their work in each daily and weekly group meeting on time to show their latest working progress and prove they work

towards the expected goals and milestones (see attachment) for review and feedback. They should improve their work according to the feedback.

2.4 The intern students should timely discuss and respond to the inquiries and messages in the group. They should also closely assist and collaborate with each other to solve any problems and deliver the expected outcomes together.

2.5 The intern students should comply with the research and development integrity and in accordance with the agreed co-authorship, copyright and intellectual property based on their participation and contribution.

3. Tasks and Duties of Intern Host

3.1 The host will check the work progress of the intern students, give feedback to student's work, and direct the outcomes of students towards the expected goals and milestones (see attachment).

3.2 The host will report the working progress and outcome of the intern student to the STEM Internship Scheme after the internship ends. The host can terminate the internship when intern students don't follow the instructions and meet the expectations of the host.

3.3 The host will try to assist the intern students to finish their work as much as possible, but it's not guaranteed and intern students need to independently finish all their work with full responsibility on their work even if no assistance is given.

3.3 The host can end the internship of the intern students at any time if the intern students have no satisfactory working progress and outcomes towards the expected goals and milestones (see attachment) of the internship.

4. Acknowledgement:

I, Mo Xianyu, confirm that the above contract terms and the following internship content are clear and I fully agree with them. I acknowledge and agree that if I fail to complete the agreed work content and requirements during the internship period, my internship will be terminated beforehand. I acknowledge and agree that if I fail to complete the agreed work content and requirements during the 28-day probationary period, my internship will be terminated in advance and I will not receive any internship salary. I acknowledge and agree that the internship salary will only be provided by the university's STEM internship program. After the internship, I will only request and receive the internship salary from the university.

My signatures for confirmation: Mo Xianyu

Internship Tasks and Requirements

Intern student should work on the following research and development work. The process and result of their work should be fully reproducible and clearly documented. Student should submit their work for verification in time. After passing the verification, they can continue working on the next task with the corresponding reward.

Intern Topic: LLM-based Proactive AI agent based on Augmented Reality (AR)

Develop an self-adaptive AI agent protocol with LLM-based AI agent that can automatically find and connect all different third-party AI agents online based on the demand of users, and facilitate their communication and collaboration to meet the demand of users. To do so, it should achieve the following features:

Part 1. Multi-modal AI agent for collection and recognition of user situation in real time

Develop a LLM-based multi-modal AI agent to analyze the collected real-time information and data from AR systems and devices including the personality, environment, and behaviors of AR users, and efficiently identity and accurately recognize the demand of users. Multi-modal LLM can accurately analyze and identify the following insights about user from the data from AR device and system:

1.a **external surroundings of user in the real world** *(eg. where: classroom, restaurant, home; who: classmate, server, mother; what: class, dining, rest)*

1.b **status and behaviors of user in the real world** *(eg, reading books, having dinner, feeling comfortable, want to open the window)*

1.c **user interactions and reactions with external surroundings** *(eg. User often listens to music when going to library and starting to study)*

1.d **personalized data and information of the user from his/her account in AR system** *(eg. living in Sichuan; likes to eat spicy food for launch)*

1.e **user interactions with AR system and applications** *(eg. User often plays and listens to music when going to library and starting to study)*

1.f **user interactions with our agent and other agents in AR system** *(eg. User often raises and accepts photo-taking requests when seeing splendid mountains)*

The actual information AI agent can get depends on the rights given by the AR system and device. After analytics and identification, multi-modal LLM should successfully recognize **which user is doing what in where and when under what circumstances**. Then it will pass the information to reasoning LLM for demand analytics and identification.

Part 4. Real-time generation of UI&UX for user perception & interaction with the solution via multi-modal AI agent

After adjusted and confirmed by the user (if needed), AI agent will execute the solutions in AR system and applications, and return the execution results back to the user. The generated solutions and execution results need to send to LLM-based multi-modal AI agent for real-time UI&UX generation. Users can perceive and interact with the solution , and they may adjust the solutions with new requests. In that case, AI agent needs to adjust and improve the solutions accordingly to generate new solutions for users' reconfirmation, and execute the new solution after reconfirmation.

4.a identify whether AI agent shows the solution first and asks for confirmation before execution, or directly executes the solution and displays the result to user *(eg. Direct execution for information search, need confirmation for shopping and payment)*

If AI agent needs to show the solution first:

4.b show the solution in an appropriate and adaptive way according to 3.d based on the situation and demand of user

4.c generate the user interaction with the solution in an appropriate and adaptive way according to 3.d based on the situation and demand of user

After AI agent executes the solution:

4.d show the execution result of the solution in an appropriate and adaptive way according to 3.d based on the situation and demand of user

4.e generate the user interaction with the execution result of the solution in an appropriate and adaptive way according to 3.d based on the situation and demand of user

After user interacts with the solution with adjustments or new requests, AI agent will execute 4 and 5 again for new solution generation and execution. It will also record and store the interaction to better understand user demands for future retrieval.

Expected timeline & outcomes:

Time	**Goal**		**Time**	**Goal**
Dec 2～5	Finish 1.a&b		**Jan 5～9**	Finish 4.b
Dec 8～12	Finish 1.c&d		**Jan 12～16**	Finish 4.c
Dec 15～19	Finish 1.e&f		**Jan 19～23**	Finish 4.d

Dec 22～26	Finish part 1 and connect part 2; begin part 4 and connect part 3		**Jan 26～30**	Finish 4.e
Dec 29～31	Finish 4.a		**Feb 2**	Final revision & delivery

Part 2. LLM-based AI agent for detection and identification of user demands in real time

According to identification results of multi-modal LLM, reasoning LLM will identify the real-time intentions and demands of the user including:

2.a What user wants *(eg. information, content, actions, and online services from the real world, our agent and other agents, AR system and applications) (eg. User is a student and in the previous class his/her teacher asked all students to make a presentation about climate change in the next class. The user made a slide with script in PowerPoint, and in class when the teacher asks him/her to present, he/she goes to the platform and will present, he needs to show the corresponding slide in classroom's projector, and view the slide with script from PowerPoint)*

2.b When user wants *(eg. timing, conditions, actions) (eg. User wants to directly show the corresponding slide for all students and teachers in class and view the corresponding slide with script to present when the teacher asks him/her to present, and the user will present in class)*

2.c Where user wants *(eg. location, environment, scene) (eg. User wants to directly show the corresponding slide for all students and teachers in class and view the corresponding slide with script to present at the platform in front of students and teachers in the classroom)*

2.d How user wants *(eg. UI&UX, audio) (eg. notification, alert) (eg. button for confirmation, multi-choice selection, gesture, text, voice)*

If no demand is identified due to limited historical data, our LLM should also observe and record the situation, demand, and solution of user, and store them into database for future retrieval and analytics. In this way, when users have the similar situation next time, our LLM can detect and identify the corresponding demand to provide the corresponding solution in no time.

Part 3. LLM AI agent for solution generation and execution for fulfilling user demands

After determining the demand of users, make the AI agent above generate and execute the corresponding optimal solutions including the content and services users need in AR, and pass the generated and executed optimal solutions with AR

content and services to the next step for real-time UI&UX generation. The generated solution should contain:

3.a **the objects AI agent will operate** *(eg. real-world items, other agents, AR system and applications)*

3.b **the actions AI agent will take on the objects above** *(eg. open, call, order, send data, retrieve result)*

3.c **the conditions AI agents will take the actions on the objects above** *(eg. When getting the confirmation of user, after the previous object finishes the execution and returns the result, after the price of the flight reduces to how much)*

The generated solutions will form a sequence of operations including the elements above. If multiple solutions are generated, AI agent will send all of them to Project 2 for users' consideration, but will also rank and select the best one for the user.